UC San Diego
EXTENDED STUDIES

Course Syllabus

Course Title:	EmbCntrlrPrg for Real-Time Sy
Instructor Name:	Vijay Kumar
Course Number:	ECE-40097
Section ID:	198192
Quarter:	SU26
Course Dates:	06/23/2026 – 08/22/2026

Welcome to Embedded controller Programming for Real-Time Systems!

I am excited to have you in this class and we will work together to demonstrate application of embedded controllers to actual event-driven system design, interrupts handling, and communication between interrupts and main thread. I will cover programming of microcontroller and peripherals, such as Timers, USART, DMA and RTC.

Communication Policy

I want to hear from you! It's not unusual to encounter difficulties with course content or assignments. If that happens to you, please reach out to me as soon as possible so that we can begin to solve the problem. The earlier you communicate with me about a potential issue, the more likely we are to find a good solution. I will do my best to respond to messages as soon as possible and within 48 hours, unless otherwise stated.

Click on 'Help' in the universal navigation menu anywhere in Canvas to visit the *Canvas Student Orientation* course to see guidelines for appropriate digital communication etiquette between students and instructors.

Course Description

This advanced programming course covers real-time event-driven applications with instant and reliable access to systems resources for embedded microcontrollers.

The design of complex electronic systems requires a firmware that will respond, within a given time, to a multitude of possible inputs, any of which may arrive at unpredictable times and in an unpredictable sequence. This problem is common in most of embedded systems. It is also very common in operating systems of modern computers.

In this class you will learn about: low-level microcontroller programming, hardware aspects, interrupt-driven programming, I/O interfacing, timers and signal conversion. The main purpose of this course is to demonstrate practical application of embedded controllers to actual event-driven system design, interrupts handling, and to tasks processing.

Course Prerequisites

ECE-40292 Embedded Systems Hardware Design and **ECE-40291 Embedded Controller Programming with Embedded C** or equivalent knowledge and experience.

Student Learning Outcomes

Student Learning Outcomes are specific and measurable goals for student learning. Students demonstrate their mastery of these outcomes through course assessments.

By the end of this course, students will be able to:

- Understand controller (Arm Coretx-M4), controller peripherals, programming the controller in C and Assembly language, Interrupts and Timers.
- Provide hands on experiences of writing interrupt service routines (ISR), and subroutines in Assembly language
- Develop practical knowledge of designing, and development of embedded software for real time embedded systems. This includes the best design practice with and without operating system.
- Learn about interrupt controller and will be able to read vector table and write interrupt service routines for GPIO and reset button.
- Learn about devices programming to use for application such as delay, counters, alarms, watchdog.
- Learn about event driven embedded design and will be able to write software based on events, e.g., character received on UART, pressing reset button on the development Kit.

If you are unclear about the outcomes for this course or their application, or if you would like more information about them, please reach out to me.

Course Materials

Required Materials. The following required materials are essential to your learning and will be actively used during this course.

Project Kit:

- Students should use Discovery Kit B-L475E-IOT01A. This is same kit that have been used in other courses.

Textbook:

- Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C: Fourth Edition

Reference:

- https://www.st.com/en/evaluation-tools/b-l475e-iot01a.html
- https://www.st.com/content/ccc/resource/technical/document/programming_manual/6c/3a/cb/e7/e4/ea/44/9b/DM00046982.pdf/files/DM00046982.pdf/jcr:content/translations/en.DM00046982.pdf
- https://developer.arm.com/documentation/
- https://www.st.com/resource/en/datasheet/stm32l475vg.pdf

You can buy your course textbooks or get information about them from the UC San Diego Bookstore. You can visit the UC San Diego Bookstore Website, call them at (800) 520-7323, or visit in person. You may also be able to buy or rent books from other online retailers.

Course Schedule

Below, please find the schedule for course topics and assignments. These dates are an estimate to help you with your planning. Please note that this schedule is subject to change during the course.

Session	Topic	Reading	Assignments w/due dates	Points
1	Introduction to Real Time Embedded system	Chapter 2	Assignment #1	7 pts
2	Microcontroller Architecture	Chapter 1, 4.1	Assignment #2	7 Pts
3	Microcontroller Instruction sets	Chapter 3, 4	Assignment #3	8 Pts
4	Introduction to assembly language	Chapter 5,6	Assignment #4	7 Pts
5	Introduction to Interrupts	Chapter 11	Mid-Term	15 Pts
6	Interrupt Programming	Chapter 11		
7	Timers	Chapter 16	Assignment #5	6 Pts
8	RTC/DMA	Chapter 16.5, 17	Final Project	30 Pts
9	Embedded System design	Chapter 7, 20	Quiz	10 Pts
	Discussion board participation		Every week	*10*
	Total possible points			*100*

Grading Policies

Letter grades are based on the UC San Diego Extended Studies Grading Scale. Your final course grade is based on the percentage of points you have earned.

Passing Grades		Non-Passing Grades	
A+	100%	**D**	60-69%
A	93-99%	**F**	59% and below
A-	90-92%		
B+	87-89%		
B	83-86%		

B-	80-82%	
C+	77-79%	
C	73-76%	
C-	70-72%	

All UC San Diego Extended Studies students must follow the UC San Diego Extended Studies Academic Integrity Policy, which oversees all instances of academic misconduct, including but not limited to: plagiarizing, cheating on exams, allowing someone access to your online course, and improper or missing citations in coursework.

For this class to count towards a certificate program, you must select either *Letter Grade* or *Pass/No Pass* as your grading option. If you select Not for Credit (NFC) as your grading option, your completion of the class will not count towards the certificate. If you'd like you change your grading option for this course, you may do so through your MyExtension portal before the final class meeting (or, for online courses, by 11:59 p.m. on the day before the published end date the course) or before final grades are posted, whichever comes first.

Weighted Grading Criteria

Your grade in this course will be weighted according to the following criteria.

Discussion Board participation:	10%
Assignments/Programming	35%
Tests/Quizzes	10%
Mid-Term and Final	45%
TOTAL	100%

Click on 'Help' in the universal navigation menu anywhere in Canvas to learn about checking your grades.

Late Work Policy

- An assignment is considered late if not submitted by due dates. No exceptions.
- Late assignments (anything posted or sent after the due date) will be graded -1 point for each day late unless advance arrangement has been made.
- Late assignments will be accepted at the discretion of the instructor and cannot be accepted more than 1 week late.
- Assignments sent with the wrong naming convention or in the wrong format will be considered late until they are sent correctly.

Assignments

Follow the naming convention below to submit assignments.

File naming/Format: As requested in the assignment.

Assignment Quality:

Coding assignment:

- Make sure code compiles and runs on the development kit.
- Comments should be sufficient and meaningful.
- Make sure to submit the code as requested in each assignment.

Discussion Board:

- I plan to have discussion board every week for this course.
- To participate, check Course Menu>Discussion Board and look for the forum for the current week.
- You are expected to maintain a regular and substantive presence on the discussion board in this course, meaning that your postings and responses to the Discussion Board prompts are geared toward expanding the learning experience for yourself and your classmates in meaningful ways.
- When I grade your Discussion Board work, I am most interested the quality of your postings/responses. There might not be right or wrong and wrong answers. Your contributions should add to the knowledge base of all of the participants in this class and provide substantive thought to receive points. If you are less knowledgeable about a topic, pose questions about the topic and/or share what you find in researching the topic yourself.

UC San Diego Extended Studies Policies and Resources

MyExtension

Your MyExtension account is your student records portal. Log into MyExtension to enroll in a course, drop a course, request verification of enrollment, request official transcripts and more.

Academic Policies and Procedures

Please refer to the UC San Diego Extended Studies Website (Student Resources tab) for specific details about academic policies and procedures. Navigate to the Grades section for grade information.

Conduct Code

All UC San Diego Extended Studies students are part of the UC San Diego community and are expected to follow University and UC-wide policies, including the Standards of Conduct and the UC San Diego Principles of Community. Reports of alleged violations involving sex offenses, including sexual assault and sexual misconduct, will be handled under the policies and procedures set forth in the University of California's Sexual Violence and Sexual Harassment Policy.

Emergencies on Campus

In the event of an emergency, information will be posted on the UC San Diego Extended Studies Website. Extended Studies students must access the website to find out the status of the emergency situation. Email and or phone lines may not be accessible. Information will be updated online as the situation progresses and an ALL CLEAR will be posted on the website once the situation is resolved.

Services for Students with Disabilities

UC San Diego Extended Studies is committed to providing equal access and an exceptional learning environment for all students. If you have any problems accessing course material, or if you anticipate or experience physical or academic barriers based on disability, we encourage you to contact our Services for Students with Disabilities Office to apply for reasonable accommodations. You can reach this office by email at unex-ssd@ucsd.edu or by calling (858) 822-1366.